



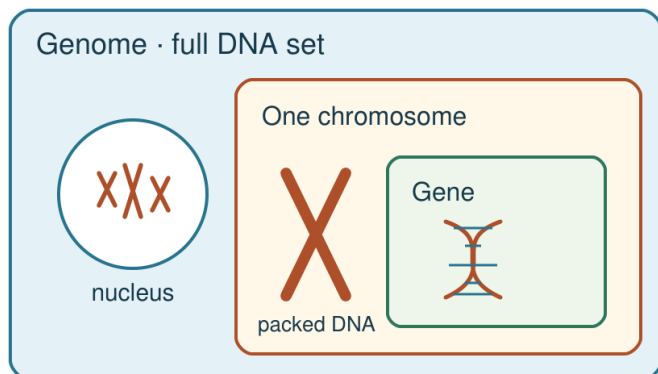
DNA, genes and chromosomes

Today I can... ☐ I can label DNA, a gene and a chromosome in a simple model. ☐ I can explain that a gene is a section of DNA with a protein instruction. ☐ I can distinguish one chromosome from the genome.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____

Arrival

Look at this week's string-and-clip DNA model. Point to one detail you notice.



My response

Starter

Recall three ideas from earlier lessons. Show one answer at a time.

1. What is one reason a claim of a new stem-cell treatment needs evidence?

2. Which structure carries the matching chromosome sets after mitosis?

3. Where are chromosomes found in a eukaryotic animal cell?



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Talk-then-sort

Complete the missing step in the part-done model. Speak, then match the cards.

DNA cord → a short marked _____ codes for a protein → chromosome package → the genome contains all DNA.

Choices: DNA · chromosome · gene · genome

Card or step	My group or match
short coloured DNA section	
whole genetic molecule	
long DNA package	
all of the organism's DNA	

Two diagrams label a gene differently. Choose the accurate model and explain one reason.

Independent F

Label a gene, DNA, chromosome and genome on the model.

This short marked section is a ...; the long package is a ...

My response

Independent M

Explain why a gene is a DNA section, not another material.

A gene is part of ... and codes for ...

My response



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Independent H

Correct a faulty gene label and explain the protein instruction.

The faulty label is ...; it should be ... because ...

My response

Go deeper

What distinctions disappear if the whole genome is called one gene?

My response

Repair two faulty diagrams: DNA inside a gene, and one chromosome as the entire genome. Justify both.

My response

Exit

Show what you know with a model and explain one reason.

My exit response

My feedback was ... Name what you heard at Review.

Next I need to ... Use “section of DNA” and “codes for a protein” precisely in one sentence.

Word help

Word	Meaning	Example	My language
DNA	the molecule that stores genetic instructions	DNA contains the instructions for making proteins.	
gene	a section of DNA that codes for a specific protein	The coloured section of the model stands for one gene.	
chromosome	a long DNA molecule packaged with proteins in a cell	A chromosome contains many DNA sections, including genes.	
genome	the entire DNA of an organism	The genome includes all of the organism's chromosomes and other DNA.	